

2. Malay: not a hard problem

Scoring: 1 point for each completely correct Malay word; no half marks. (max 24)

- Accept *itu* 'the' after *Kecantikan* in a, showing it's generic.

a	Kecantikan bukan hadiah.
b	Gadis kaya itu bukan penyani.
c	Kekayaannya bukan untuk gadis itu.
d	Manusia itu tidak datang.
e	Hadiah itu dari penyani itu tidak cantik.

Comment

- Abstract nouns ("beauty") are formed by surrounding the respective adjective (beautiful) with *ke~an*.
- Words that modify a noun, such as adjectives and *itu* ('the') follow it, and *nyai* 'his' is a suffix.
- Malay has two verbs translating 'is not': *tidak* is used when followed by an adjective or verb, *bukan* is used otherwise.

1. Tocharian

Scoring (max 39)

- **2.1.** 1 point for each correctly ordered stage (assume a stage is correct unless it is located later than a stage which it should precede) (max 7)
- **2.1.** 1 point for each correctly stated change linked to the correct stage. (max 6)
 - Only accept maximally general rules – e.g. ‘j > g’ but not ‘j > g/_o’
 - Only accept maximally simple rules – e.g. ‘b^h > 0/m_’, but not ‘mb^h > m’.
 - Accept plausible alternatives, e.g. ‘s > 0/o_’ or ‘b^h > 0/ not at beginning of word’.
- **2.2.**
 - 2 points for each correct ordering statement; 1 for undergeneralised statements. (max 16)
 - 1 point for each correct justification. (max 8)
 - 2 points for a diagram showing global ordering (max 2)

Q.2.1

order	A-G	Change(s) that turned this stage into the next one
1 (earliest)	D	c > k
2	C	j > g
3	G	b ^h > Ø / m _
4	F	b ^h > p
5	B	g > k
6	A	s > Ø / _ #
7 (latest)	E	

Q.2.2 [Please ask for extra paper if you need it.]

The rules given in the question are repeated below.

1	o > ë	10	m > n / _ t
2	nt > Ø / _ #	11	u > ə
3	e > ə	12	k > ś / _ e, ē
4	or > ur / _ #	13	w > w ^y / _ i, ī
5	ē, ī, ū > e, i, u	14	g > k
6	m > n / _ #	15	d > Ø / _ #
7	t > c / _ e, ē	16	s > Ø / _ #
8	d > ś / _ e, ē	17	ti > Ø / n _ #
9	n > Ø / _ #		

The first thing to realize here is that (as the question suggests) only a partial order can be deduced. For example, 15 can be put anywhere in the ordering, because it doesn’t have an effect on any other rules. The rules that can be ordered are the ones that make reference to each other’s inputs and outputs.

One useful notation for ordering changes is A + B, meaning ‘change A applied before change B’, but other notations are of course possible. It would be tempting, but confusing, to use ‘>’ here as well as within the rule.

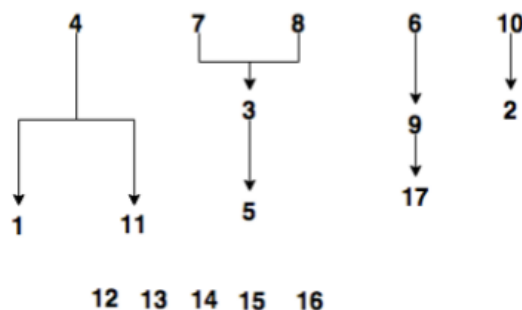
The orderings it is possible to work out (by a kind of modus tollens) are given below.

- 4 applied before both 1 and 11 (4 + 1, 11) [two orderings]

In *akëntər*, we have: or > ur (by 4) > ər (by 11). If the order were 1 + 4 (meaning ‘1 before 4’), we would have: or > ər (by 1) and 4 would fail to apply. If the order were 11 + 4, we would have: or > ur by 4, but then no subsequent change to ər because 11 is no longer available.

- 7 and 8 both applied before 3 (7, 8 + 3) [two orderings]
In *cək^wē*, we have: té > cé (by 7) > cǎ (by 3). If the order were 3 + 7, we would have: té > tá (by 3) and 7 would fail to apply. The same reasoning applies to 8 in the word *śákə* (‘ten’).
- 3 applied before 5 (3 + 5)
In *pacér*, we have: é̇ > é (by 5), but 3 is no longer available to change é to á. If the order were 5 > 3, we would have: é̇ > é (by 5) > á (by 3).
Also *kēruwos* > *ser*...
- 6 applied before 9
In *kántē*, we have: m > n (by 6) > Ø (by 9). If the order were 9 > 6, we would have: -m > -n (by 6), but then no subsequent deletion because 9 is no longer available.
- 10 applied before 2
In *śákə*, we have: əmt > ənt (by 10) > ə (by 2). If the order were 2 > 10, we would have: əmt > ənt (by 10), but then no subsequent deletion of nt because 2 is no longer available.
Or: *dekəmt* > *dekənt* > *səkəmt*
- 9 applied before 17
In *akēn*, we have: nti > n (by 17), where 9 has already applied earlier. If the order were 17 > 9, we would have: nti > n (by 17) > Ø (by 9).
- 12, 13, 14, 15, and 16 are unorderable with respect to any other rules.

A neat diagram of a solution might look like this, where a downward arrow means ‘precedes’.



Comment

Working out an order of the steps relative to one another should be straightforward by any method, e.g. by observing that the first word begins either b^h- or p-, and that the simplest conclusion is that this represents a historical change from one to the other (so all the forms with b^h- will be at ‘one end’ of the timeline).

This will give an ordering of the elements relative to one another (e.g. “this stage goes in between these other two”), but the next step is to tell which part is the beginning and which is the end. The trick to this is to realize that some changes are irreversible, specifically the mergers of c with already existent k (this is why the word *kóro-* is included) and later g with k. The statement in the rubric that sound changes must be regular means that the reverse of this can’t happen; that would require an irregular change of k to g in some words but not in others. (A logical possibility is that k becomes g before on but not or; candidates should realize that this is an implausible analysis.)

Scoring (max 21)

- 3.1 1 point for each correct cell (max 10)
- 3.2 1 point (max 1)
- 3.3 1 point for each correct cell (max 10)
 - accept kira or kiria
- Try not to penalise the same error twice.

Q.3.1	A	B	C	D	E	F	G	H	I	J
	36	1	81	25	16	9	4	49	64	100
Q.3.2	a = 116									

Q.3.3	2	<u>kiria</u>	4	maria
	6	waragaria	7	karia
	22	nguira-ni karia		
	44	ngui ki, ngui tebone-gonaga deria		
	66	ngui ma, ngui dauni-gonaga waragaria		
	77	ngui dau, ngui waragane-gonaga kiria		
	88	ngui dau, ngui waragane-gonaga haleria		
	173	ngui bea, ngui hombeane-gonaga <u>haliria</u> (accept halira)		

ngui bea, ngui hombeane-gonaga halira

Comment

1	mbira	9	dira
2	<u>kiria</u> ~ (ngui) ki	10	pira
3	(ngui) tebo ~ tebone(-gonaga)	11	bearia
4	maria ~ (ngui) ma ~ mane(-gonaga)	12	hombearia
5	(ngui) dau ~ dauni(-gonaga)	13	<u>haleria</u>
6	waragaria ~ (ngui) waraga ~ waragane(-gonaga)	14	deria
7	(ngui) ka ~ kane(-gonaga)	15	nguira-ni
8	<u>haliria</u> ~ haline(-gonaga)		

Scoring – score Devanagari script generously (max 26)

- **4.1** 1 point for each correct cell (max 14)
- **4.2** 1 point (max 1)
- **4.3** 1 for (a), 2 for (b). (max 3)
 - 1 for all but one letter correct in (b)
- **4.4** 1 for each correct cell. No half marks. (max 8)
 - for ta, accept a single hook (backwards h)

Q.4.1	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	G	K	A	F	-	B	J	E	L	H	M	N	C	I

Q.4.2	‘a’													
Q.4.3	(a)	paanaa					(b)	सानो						

Q.4.4	Devanagari	Roman	Devanagari	Roman
	रे	re	साख	saakha (accept 'saakh')
	सोली	solii	लक्का	lakkaa
	ल x	ta	सेनो x	seno
	पनिर x	panir	निरन्तर x	nirantar

Scoring (max 35)

- 5.1 1 point for each correct cell. (max 7)
- 5.2 1 point for each correct cell. (max 18)
- 5.3 10 points for a really clear and well argued answer. (max 10)

		Alan	Barbara	Chris
Q.5.1	octet	<u>ocamatet</u>	<u>octematet</u>	<i>i dunno...</i>
	purple	<u>purpamaple</u>	<u>purplemaple</u>	<u>purrrmaple</u>
	tuba	<u>tubamaba</u>	<u>tubamaba</u>	<u>tuuumaba</u>

Q.5.2	antiseptic	<u>antimaseptic</u>	<u>antimaseptic</u>	<u>antimaseptic</u>
	Canada	<u>Canamada</u>	<u>Canamada</u>	<u>Canamada</u>
	feudalism	<u>feudamalism</u>	<u>feudamalism</u>	<u>feudamalism</u>
	optics	<u>opamatics</u>	<u>optimatics</u>	<i>i dunno; (oppmatics also permitted)</i>
	party	<u>partamaty</u>	<u>partymaty</u>	<u>paarrmaty</u> or <u>paarty</u>
	table	<u>tabamable</u>	<u>tablemable</u>	<u>taamable</u>
	water	<u>watamater</u>	<u>watermater</u>	<u>waaamater</u>

Q.5.3

Various kinds of answers would be acceptable here; what we're looking for is nontrivial insight into what might be going on rather than an answer according to what a phonologist might answer.

Nonetheless, a top-quality answer should probably bring up the idea that the respondents are trying to achieve contradictory goals, and that different respondents' strategies here are prioritizing different goals. The important parts are these:

- They all make a single word sound like two words by
 - maintaining the original stressed vowel (V')
 - adding word stress to another of the original vowels (V'+):
 - a preceding one that already has secondary stress (Alabama, congratulations, hypothermia), and which therefore is not adjacent to V'.
 - otherwise, the next vowel but one after V' (capital, octagon, wonderuful).
 - adding *ma* just before the second stressed vowel, whether this was originally stressed or newly stressed.
- These rules presuppose at least three syllables in the original word, so they need supplementary rules for bisyllabic words (V1 [C-] C V2):
- They all add stress to V2: V1'....V2'.
- Alan produces *Cama* before C:
 - ama* between C- and C if possible (capamatain),
 - otherwise: *C*ama*, where C* = C (obamaboe).
- Barbara
 - repeats the final syllable before *ma*: *C*V*ma*, where CV = CV2
 - adding this before C (captamatain, octematet, oboemaboe).
- Chris
 - lengthens the first syllable and adds *ma*.
 - So if lengthening isn't possible (e.g. if the vowel is short and the consonant is a stop) he has no strategy.